



Vojtěch Špaček

Senior C++ engineer — safety-critical embedded & HIL test automation | Linux/networking | agentic-AI (MCP) tooling

Pragmatic programmer and creative problem solver — systems, verification and open source.

Location: Brno, CZ | Citizenship: EU (Czech Republic)

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YouTube: youtube.com/playlist?list=PLaWOvDBjg6WiUcQm-yEP1RskMfPeWMKTL | Discord:

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Profile Summary

Senior C++ engineer working on safety-critical and real-time systems: aircraft onboard diagnostics at Honeywell, electron-microscopy detector software at Thermo Fisher Scientific, and a Linux inspection proxy and firewall at Trusted Network Solutions.

Most recently sole owner of the hardware-in-the-loop verification suite and automated bench for a safety-certified two-stage gas furnace ignition controller (UL/CSA, ANSI Z21.20, NFPA 54), raising measured conformance to 259 of 271 tracked state transitions (95.6%).

Linux, networking and security specialist — Netlink, nftables, TLS inspection, Kerberos/LDAP/OCSP — with C++20 daemons shipped as Debian packages and a Rust reimplementations published on crates.io.

Builds agentic-AI tooling on the Model Context Protocol: an agent harness adopted into a daily client delivery workflow, and MCP-Prompts, published to npm, Docker Hub and the MCP directories (Glama, MCP Market, PulseMCP) with 97 stars and 21 forks.

Preferred Roles

- **Test Automation / Verification Engineer (HIL, safety-critical) — Hardware-in-the-loop benches, requirement traceability, instrument control and test evidence reporting.**
- **Senior C++ Engineer — embedded, RTOS and real-time systems — State-machine engines, hardware abstraction layers, inter-process communication, high availability.**
- **Linux Systems, Networking and Security Engineer — TCP/IP, Netlink, nftables, encrypted traffic inspection, daemon design and packaging.**
- **AI/LLM Integration Engineer — MCP servers and agentic tooling — Model Context Protocol servers, prompt and agent workflow design, LLM provider APIs, CI/CD for AI-assisted delivery.**

Work Experience

Test Automation Engineer (Python, C#)

Resideo (Ademco) (via CoolPeople (contractor))

Hardware-in-the-loop verification of a safety-certified two-stage gas furnace ignition controller (UL/CSA, ANSI Z21.20, NFPA 54), as sole owner of the verification test suite and of the automated bench that drives it. Owned delivery end to end: requirement traceability, issue tracking, branch and pull-request workflow, Conan packaging, CI integration and monthly client reporting.

- **Engineered the bench device layer so each instrument is owned by a single thread against a shared monotonic clock, covering a pyvisa/USBTMC/SCPI programmable AC source, CAN serial-bus monitoring and Modbus relay and probe I/O.**
- **Raised measured conformance to 259 of 271 tracked transitions (95.6%) and closed a ~148-case gap in which stimulus windows were never actually measured, bringing it to zero.**
- **Investigated 25+ firmware and harness defects against the C/C++ firmware sources, shipping a standalone reproduction script for each.**

Software Developer (C++)

Thermo Fisher Scientific (via Ixperta (contractor)) · 2023 – 2024

Electron microscopy software for high-resolution detector hardware operation and image acquisition. Built server-side C++ applications, FPGA-based data processing pipelines with gRPC networking, and backend logic for a C#/ .NET WPF image processing and display application. Worked in an international Agile team with bi-weekly stakeholder demos.

- **Implemented server-side software in C++, including hardware abstraction layers (HAL) and inter-process communication (IPC) using COM and MIDL on Windows.**
- **Developed embedded software for FPGA hardware, using gRPC network communication in C++, Python and Linux environments.**
- **Optimized the data acquisition pipeline and implemented robust error handling.**

Software Engineer II

Honeywell · 2022 – 2023

Onboard Maintenance System that monitors the condition of aircraft and diagnoses issues during flight. Architected and implemented the DiagTest framework, a state machine-based diagnostic test engine in C++ managing 15+ operational states for aircraft member system testing during flight and maintenance operations.

- **Designed a job-based execution system with DiagTestJob and DiagTestMsgQueProc for asynchronous test orchestration, implementing message queue processing patterns for reliable real-time diagnostics.**
- **Engineered a test interference management system preventing conflicting diagnostic tests from running simultaneously, ensuring aircraft system safety and data integrity.**
- **Implemented four diagnostic test modes (data collection, passive monitoring, interactive, non-interactive) with timeout management, precondition handling and aviation-standard command/response protocols.**

Application Developer

Trusted Network Solutions · 2020 – 2022

Kernun Adaptive Firewall. Implemented a forward proxy in C++ on Linux featuring HTTP/HTTPS proxying with TLS/SSL inspection, OCSP integration for certificate validation, Kerberos authentication and LDAP user/group lookup, plus the ipmon network monitoring daemon driving nftables firewall rules from real-time IPv4/IPv6 address changes.

- **Built comprehensive proxy infrastructure with certificate generation and management (CA, client certificates), a dynamic configuration builder/parser system, and a manager/partition architecture for modular daemon design.**
- **Integrated security components including Suricata (intrusion detection) and nftables (static firewall), aggregating logs and managing connection information in PostgreSQL.**
- **Developed DNS resolution and caching mechanisms, domain categorization and client authentication methods.**

Independent IT Contractor

Self-employed

Independent contracting for business and consumer clients: CI/CD automation pipelines and Linux server administration for B2B customers, and IT support, consumer hardware retail and security consulting for B2C customers.

- **B2B: CI/CD automation pipelines and Linux server administration.**
- **B2C: IT support freelancing, consumer hardware retail and security consulting.**

Availability & Work Authorization

Available as an independent contractor (Czech IČO). Brno on-site and remote. Trading on a Czech IČO. EU citizen (Czech Republic) — no permit required in the EU/EEA.

Languages

Czech (native) · Slovak (native) · English (C1) · German (A1)